Manuscript outline

Revised Figures 2–4: mechanisms, geometry and polarity dynamics

Main figures, legends and selected supporting analyses

19 October 2026 · Current Figures 2–4 candidates for author discussion

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

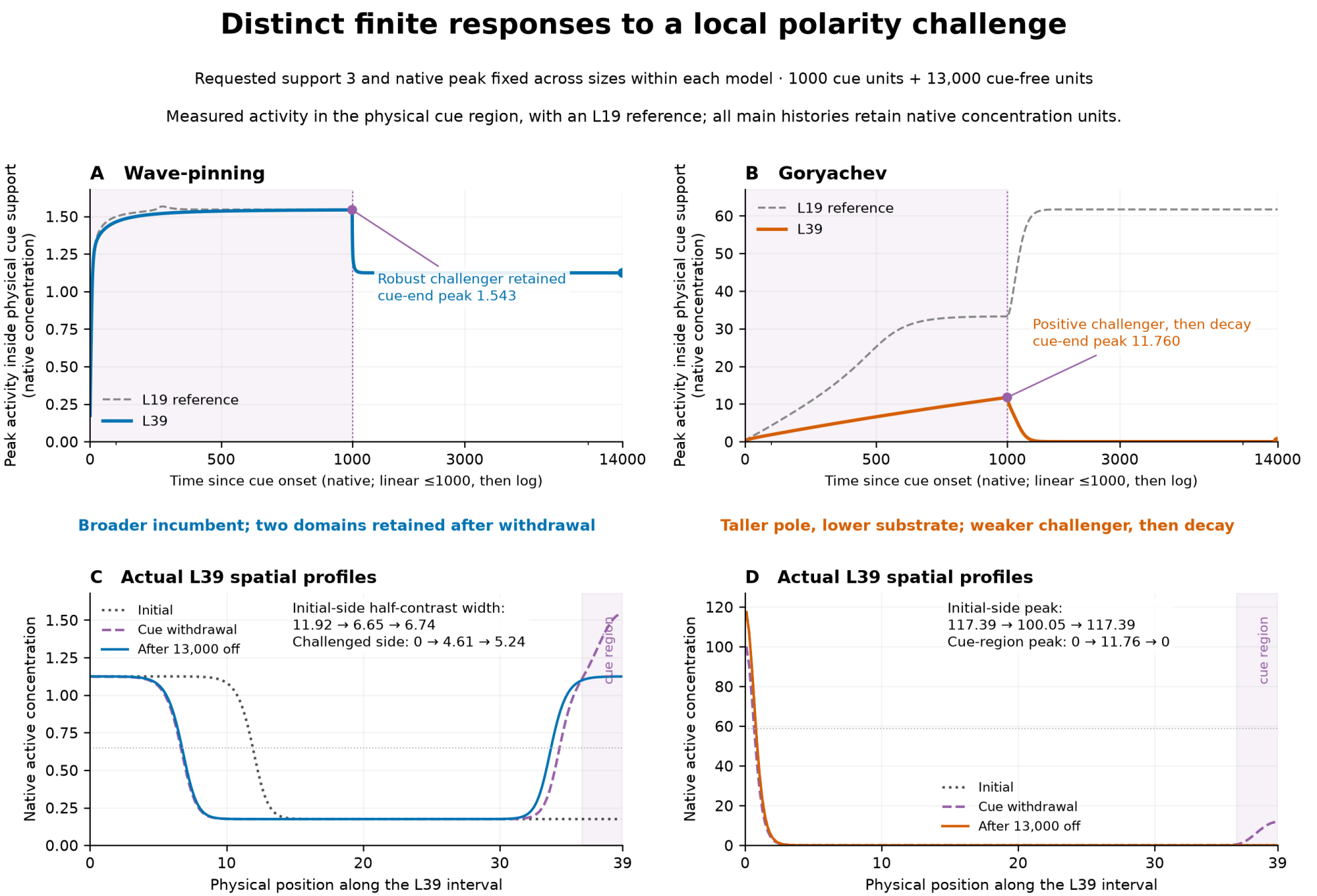
Figures 2–4 focus on three questions: which measurements separate explanations for difficult polarity replacement; how local activity becomes global polarity on a ring; and how physical site motion, pattern organization and geometry contribute to angular memory. Retained simulations and targeted distance–duration and Spring-ring experiments supply the evidence. Figure 2 now uses the separate finite-response candidate and Figure 3 the completed three-size stationary-cue review. Their emphasis and main composition remain provisional. Detailed assay maps and secondary diagnostics are supporting material.

This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

Proposed main figures

- Figure 2. Distinct finite responses to local polarity challenges
- Figure 3. Local activity and retained orientation under stationary cues
- Figure 4 · Directional memory and physical site dynamics

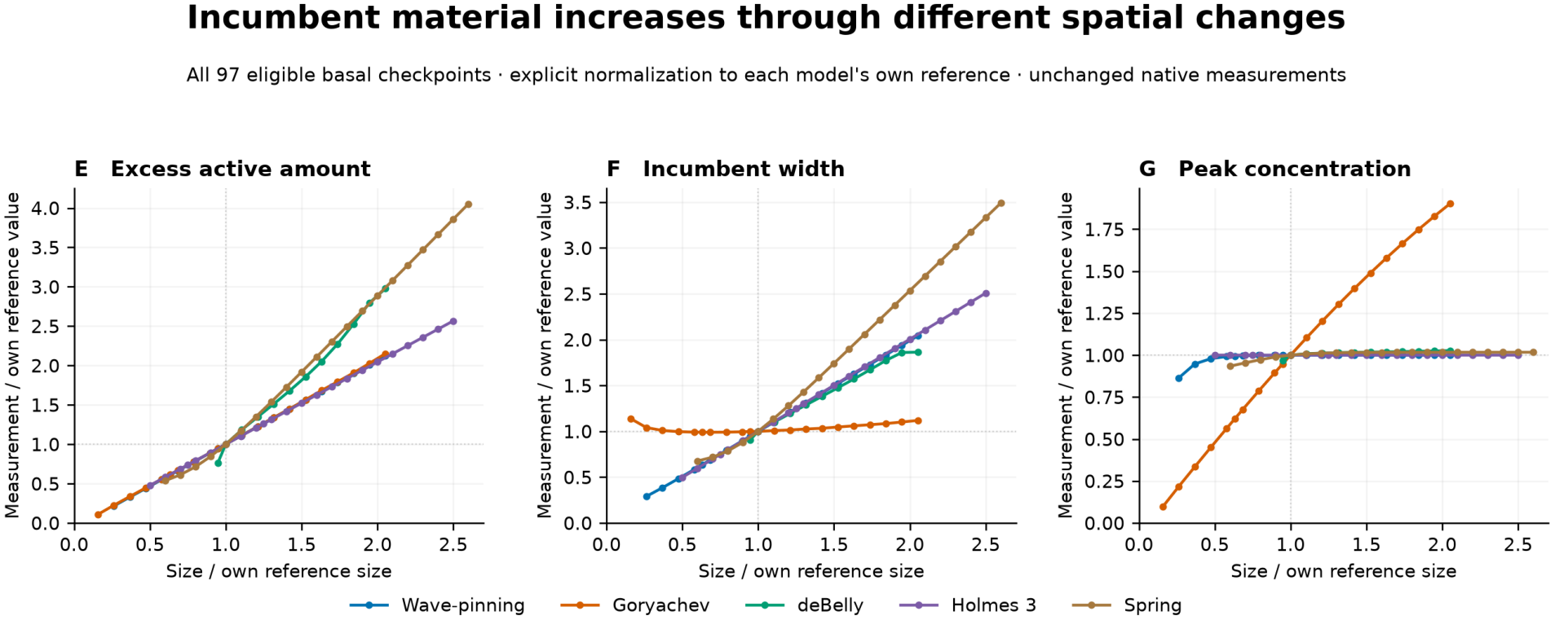
Figure 2. Finite challenge responses · image 1 of 2



Both native inputs transfer stimulus × inactive into active; finite responses include coupled reaction and transport.
Native peaks: WP 0.1; Goryachev 6.4. These are not matched optical doses or biological clocks.
Widths use each initial half-contrast threshold; zero threshold-defined extent does not mean absent activity. Finite patterns are not tracked site identities or a movement law.

Finite native activation and withdrawal responses

Figure 2 continued. Incumbent organization · image 2 of 2



All-five incumbent organization

Figure 2. Distinct finite responses to local polarity challenges

Similar size-dependent reversal behavior can arise through distinct mechanisms, which predict different observable responses to polarity challenges.

Detailed legend

A–B: Actual peak activity in the physical right-hand cue footprint [L−3,L] at L19 (gray reference) and L39 (model colour). Each model uses the same requested physical support 3, peak and 1000-unit duration across those sizes: WP 0.1, Goryachev 6.4, followed by 13000 cue-free native units. Both inputs add stimulus×inactive to active, with equal loss from inactive. Time is linear through 1000 then logarithmic, with withdrawal marked at 1000; the final coordinate 14000 is 13000 after withdrawal. Concentrations are native and are never independently peak-normalized. Goryachev L39 develops a positive peak 11.7598; it is 10.581 at 10 off and 5.2796 at 100 off, then approximately zero by 1000 off. Its L19 reference instead grows to 61.676 after an initial small withdrawal dip. WP's two sizes have nearly equal challenger height, including cue-end peaks 1.5460/1.5434, despite different remaining incumbents.

C–D: Exact L39 spatial fields initially, at withdrawal, and after 13000 cue-free units. Purple shading is the physical cue footprint, not activity extent. Dotted horizontal lines give each case's initial background-plus-half-range threshold. WP39 retains two connected above-threshold domains in every saved off-frame (2027 samples); their final initial-side/challenged-side widths are 6.7382/5.2372. Goryachev L39 never forms a challenger above its initial half-contrast 58.695, but its positive cue-driven activity is explicitly visible in B and in the withdrawal profile. Its initial-side peak recovers from 100.052 at withdrawal to 117.390. Profile values are measured fields, not evidence inferred from a scalar-strength gate. The endpoint two-field comparisons in the source table remain distinct from any scalar criterion. Connected regions refer to spatial patterns, not tracked identities or protrusions.

E–G: All 97 retained basal checkpoints, with excess amount, connected half-contrast width and peak concentration divided by the value at each model's own reference size. Sizes use L/Lref (WP/Goryachev/deBelly Lref 19; Holmes 20; Spring resting L_0ref 1). Amount is the one-dimensional excess-concentration integral within the basal half-contrast core, not calibrated molecule number. Spring uses its actual physical domain for widths/amounts; deBelly retains nominal chemical-solver coordinates. Thirteen ineligible preparations remain in the linked table, without artificial zero values. At L19→39, WP's excess amount/width/peak folds are 2.124/2.042/1.000, versus Goryachev 2.152/1.119/1.903. All five curves and the original full native-unit organization artwork remain accessible.

The labels linking organization to response name conditional model predictions, not unique molecular identification or a causal allocation of reversal time. The native channels, doses, spatial discretization and clocks are not optical-actuator or reporter calibrations. The chemical grid has 201 cells at both sizes; the sampled input peak is 0.999388/0.997422 of the requested peak at L19/L39 (about 0.2% mismatch), while its

Figure 2. Distinct finite responses to local polarity challenges · continued

physical integral is nearly unchanged. This small cue-sampling mismatch does not bound basal or other discretization errors; Goryachev L39's initial core has four sample centres and spatial convergence is not established by the retained checks. Thirteen thousand units of off-follow-up do not prove an asymptotic attractor. No velocity law or cell displacement is inferred. Small-loss recovery, analytic gain and evolved interventions remain supporting evidence, not the experimental headline.

Narrative points and conclusions

- Incumbent growth takes different spatial forms: most models widen; Goryachev grows taller and lowers available inactive substrate.
- Under the native stimulus-times-inactive channel, Goryachev has weaker size-dependent challenger activity; its L39 response is positive and later decays. WP has nearly size-invariant challenger height but a larger remaining incumbent domain at L39.
- After withdrawal WP39 retains two observed active domains while Goryachev39 restores the initial-side pattern over the finite follow-up. These are native response routes, not unique mechanistic classifications.

Proposed measurements and experimental follow-up

Activity peak/integrated increment inside the actual physical cue support, with native scale and a fixed within-model cue schedule.

Incumbent concentration, half-contrast extent/edge retreat and activity beyond the cue, followed after withdrawal.

Basal excess amount, connected width and peak concentration, preserving all97 native checkpoints and explicit reference normalization.

Scope, limitations and pending validation

Native stimulus channel, dose and timescale are not experimentally calibrated; no optical actuator or reporter model is selected.

Finite 13000-unit off observations do not establish asymptotic stability, site ancestry, whole-field replacement from a scalar gate, or a movement law.

Goryachev39 has four native sample centres in its incumbent core; the retained evidence is not a spatial-convergence demonstration.

Observed response signatures and organization do not uniquely identify molecular mechanisms or causally partition reversal resistance.

Figure 2. Distinct finite responses to local polarity challenges · continued

Sources and reproducibility

Methods, exact source arrays and checks
Native protocols and endpoint comparisons
Original unchanged organization artwork

Figure 3 review: constructing the activity vector · image 1 of 3

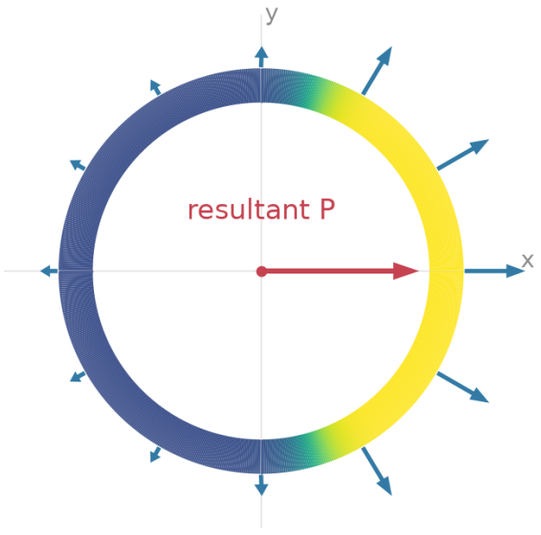
Two activity peaks can leave a weak resultant

Activity u is real; x + iy stores its weighted normal's two coordinates.

$$M_1 = \sum_j u_j \Delta s_j (\cos\theta_j + i\sin\theta_j), \qquad M_0 = \sum_j u_j \Delta s_j$$

$P = M_1/M_1(0)$ rotates and scales the initial resultant to 1.

A Initial basal activity



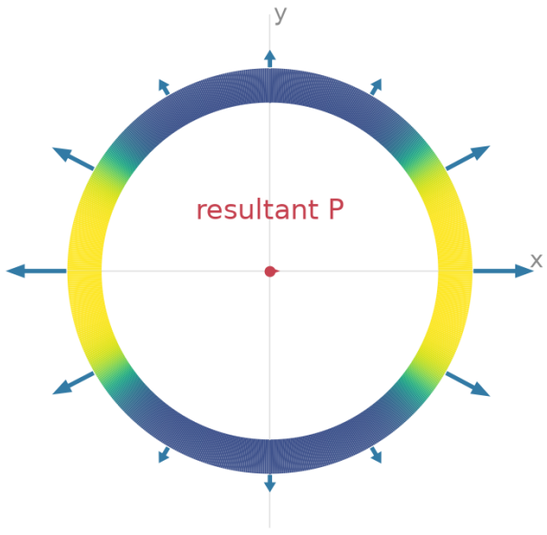
Before the antipodal cue

Total activity $M_0/M_0(0) = 1.0000$

Resultant magnitude $R = |P| = 1.0000$

Initial normalization sets P(0) = 1.

B Retained endpoint: opposing peaks



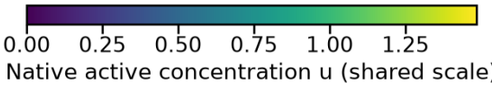
1000 cue units + 13,000 cue-free units

Total activity $M_0/M_0(0) = 0.9975$

Resultant magnitude $R = |P| = 0.0399$

Heading is not interpreted when R < 0.1.

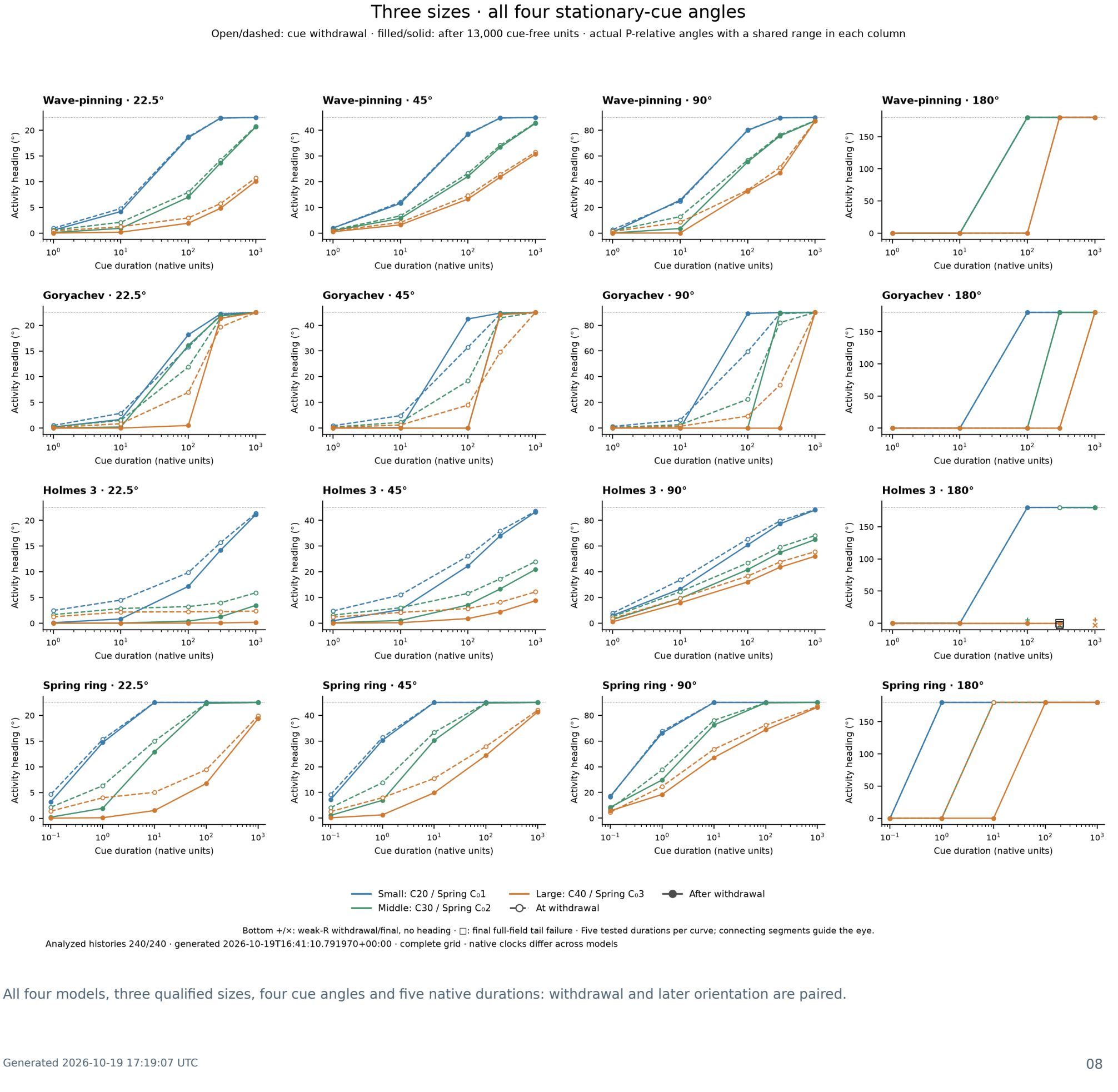
Blue arrows: exact sums of native cell contributions in 30° sectors.



All arrows use one vector scale.
Holmes 3 · fixed circumference C = 40.

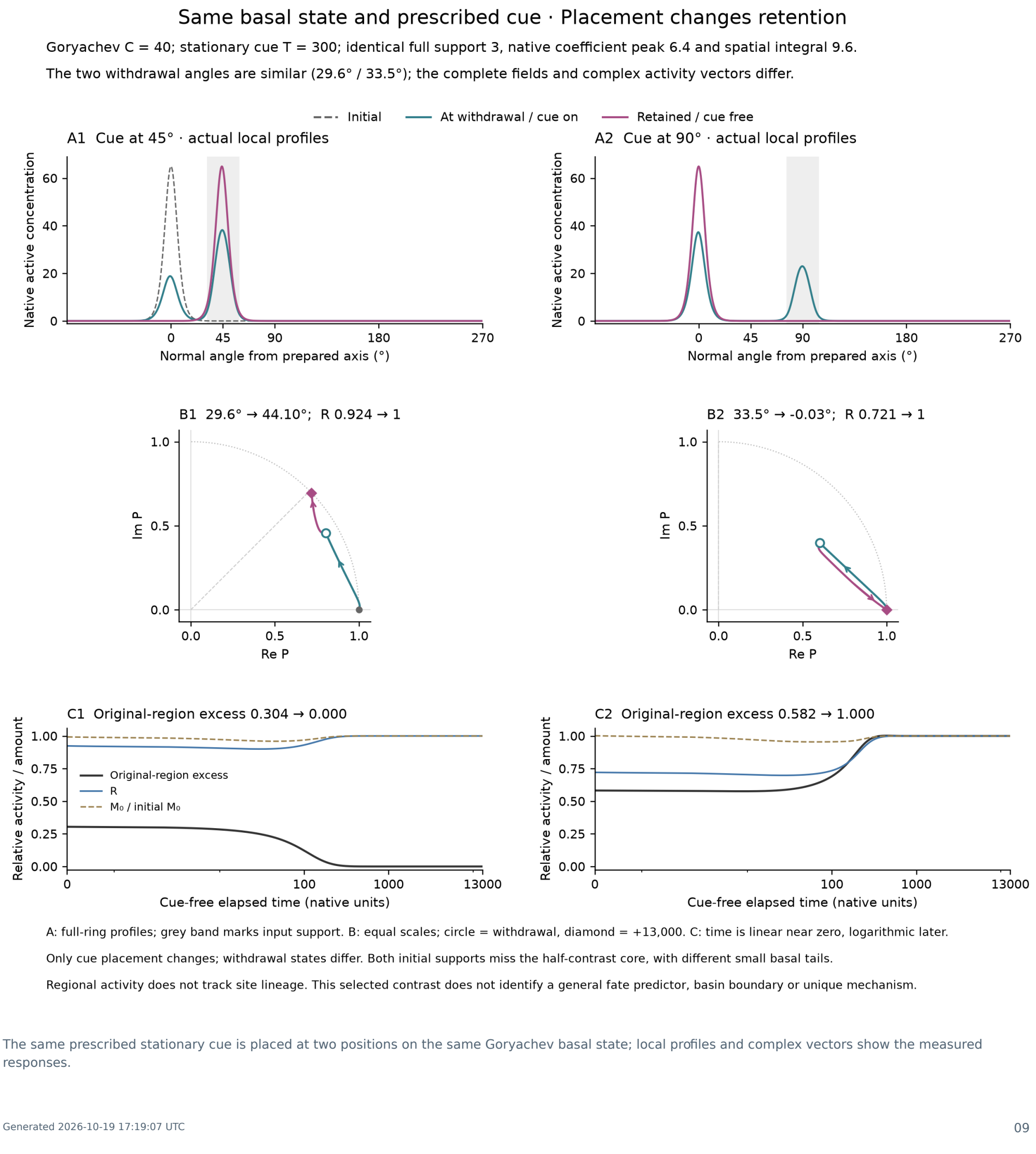
Measured Holmes profiles illustrate the physical vector sum and the distinction between resultant magnitude and total activity.

Figure 3 review: the complete stationary-cue experiment · image 2 of 3



All four models, three qualified sizes, four cue angles and five native durations: withdrawal and later orientation are paired.

Figure 3 review: a controlled placement contrast · image 3 of 3



The same prescribed stationary cue is placed at two positions on the same Goryachev basal state; local profiles and complex vectors show the measured responses.

Figure 3. Local activity and retained orientation under stationary cues

One stationary-cue experiment links spatial activity to its global vector, then tests how cue placement, duration and model size affect response and retention. The following sheets are current evidence for choosing a cohesive main figure.

Detailed legend

Vector construction. Each boundary segment contributes its real active concentration multiplied by physical segment length and its outward unit normal. The complex first moment stores the summed x and y components. Dividing by its initial complex value rotates and scales the initial vector to one. Its angle describes activity orientation; its magnitude R describes the resultant relative to its initial magnitude. Total active amount M0 is measured separately. In the measured Holmes C40 antipodal example, final total activity is 0.99746 of initial while R is 0.03993: two opposing peaks nearly cancel. Blue arrows group exact native cell contributions into twelve sectors using the same vector scale as the red sum.

Experimental design. Every condition restarts an exact qualified basal state. Chemical circumferences are 20, 30 and 40; Spring resting circumferences are 1, 2 and 3. Cue centres lie at pi/8, pi/4, pi/2 or pi relative to the declared preparation axis. The frozen measured initial first-moment phase defines the small exact offset in plotted P coordinates; centres are never snapped to a different cell. Chemical cue durations are 1, 10, 100, 300 and 1000; Spring durations are 0.1, 1, 10, 100 and 1000. Each is followed by 13000 cue-free native time units.

Complete response map. Rows identify models, columns cue angles and colours the three sizes. Open/dashed curves show withdrawal; filled/solid curves show the later endpoint. Actual angles require R at least 0.1. Weak-resultant endpoints are explicitly marked without assigning them a heading. The curve joins guide the eye between five tested durations. All 240 conditions and all 125 initial cue/core overlaps are retained. The fixed physical full support is 3 for chemical models and 0.2 for Spring; native peaks are 0.1, 6.4, 1 and 4 for Wave-pinning, Goryachev, Holmes and Spring. Clocks and doses are not calibrated across models.

Controlled placement contrast. Goryachev C40 receives a 300-unit stationary cue at 45 or 90 degrees. Both conditions share the full basal state, kinetic parameters and cue shape, strength and duration. The cue arrays differ by an exact 100-cell cyclic shift; both have integral 9.6 and coefficient peak 6.4. These match the prescribed input coefficient; the native activation term also multiplies local inactive substrate, so equal activated biochemical amount is not imposed. Withdrawal headings are 29.59 and 33.48 degrees, but relative resultant magnitudes are 0.924 and 0.721 and both profiles contain two resolved peaks. After release, the 45-degree case progresses to 44.10 degrees while the 90-degree case returns to -0.028 degrees. The original material-region excess approaches 0 and 1, respectively. These are different complex vectors and local states at withdrawal.

Figure 3. Local activity and retained orientation under stationary cues · continued

Narrative points and conclusions

- Ring measurements distinguish graded retained reorientation from nearly cancelling activity. Across all 180 non-antipodal conditions, the final field is a shifted copy of its own basal pattern with one resolved activity region and recovered resultant magnitude. Intermediate angles therefore reflect a retained spatial shift rather than a weak final activity pattern.
- The finite-cue response depends on size within each native protocol. For a 90-degree cue of duration 100, Wave-pinning retains 79.86, 55.36 and 32.52 degrees across circumferences 20, 30 and 40; Goryachev retains 89.09, -0.014 and -0.003 degrees. The complete map preserves other angles and durations rather than reducing them to an arbitrary effective-duration threshold.
- Cue-on deflection and later orientation are distinct observations. The same-basal placement pair demonstrates how similar withdrawal angles can accompany different resultant magnitudes, local profiles and subsequent activity redistribution. This supports examining the activity vector and spatial pattern together; it does not identify a unique competition mechanism.

Proposed measurements and experimental follow-up

For every condition retain activity direction and resultant magnitude during the cue and release, total active amount, actual local profiles and the fixed original material-region excess. The rich tables include sampled minima/maxima, rebound, and first/retained threshold-entry brackets with finite-horizon censoring. Regional depletion can result from translation, redistribution or local chemistry and is not a site-ancestry measurement.

A boundary activity recording could supply the same weighted sum using measured boundary elements and their actual outward normals. The circle provides those normals exactly in the model. Relating this descriptor to cell movement would require an additional empirical measurement; movement is not inferred here.

Scope, limitations and pending validation

The size series keeps physical cue support and native parameters fixed. Physical cue separation, angular footprint, overlap, basal architecture and material totals consequently change with size. These experiments measure that declared protocol response rather than an isolated causal contribution of geometry.

The separate finite-tail field gate passes 228 conditions, is unavailable for 10 compact reused histories and fails for Holmes antipodal duration 300 at circumferences 30 and 40. Values and flags remain displayed. The second weak final resultant is Holmes C30/180 degrees/duration300 (R about 0.0997); the C40/180 degrees/duration1000 teaching

Figure 3. Local activity and retained orientation under stationary cues · continued

example has R about 0.0399. A finite release does not prove asymptotic stability. Repeated samples from one deterministic history are not independent replicates.

The figure composition remains under author discussion. These review sheets preserve the complete systematic evidence and a controlled illustrative pair; they are not an accepted final main-panel layout. The integrated outline assembles these provisional review sheets; the preceding prototype is preserved separately.

Model definitions and readouts

$$M_1(t) = \sum_j u_j(t)\,\Delta s_j(t)\,[\cos\theta_j + i\sin\theta_j] \tag{3.1}$$

The activity field u is real. The real and imaginary sums are Cartesian components of the physical activity vector. On each uniformly sampled circle, segment length is actual circumference divided by cell count, including Spring C(t).

$$P(t) = \frac{M_1(t)}{M_1(0)}, \qquad R(t) = |P(t)|, \qquad \phi(t) = \arg P(t) \tag{3.2}$$

The initial normalization is applied once; it aligns the initial axis and sets the initial magnitude to one. Heading is displayed only above the declared relative-resultant gate.

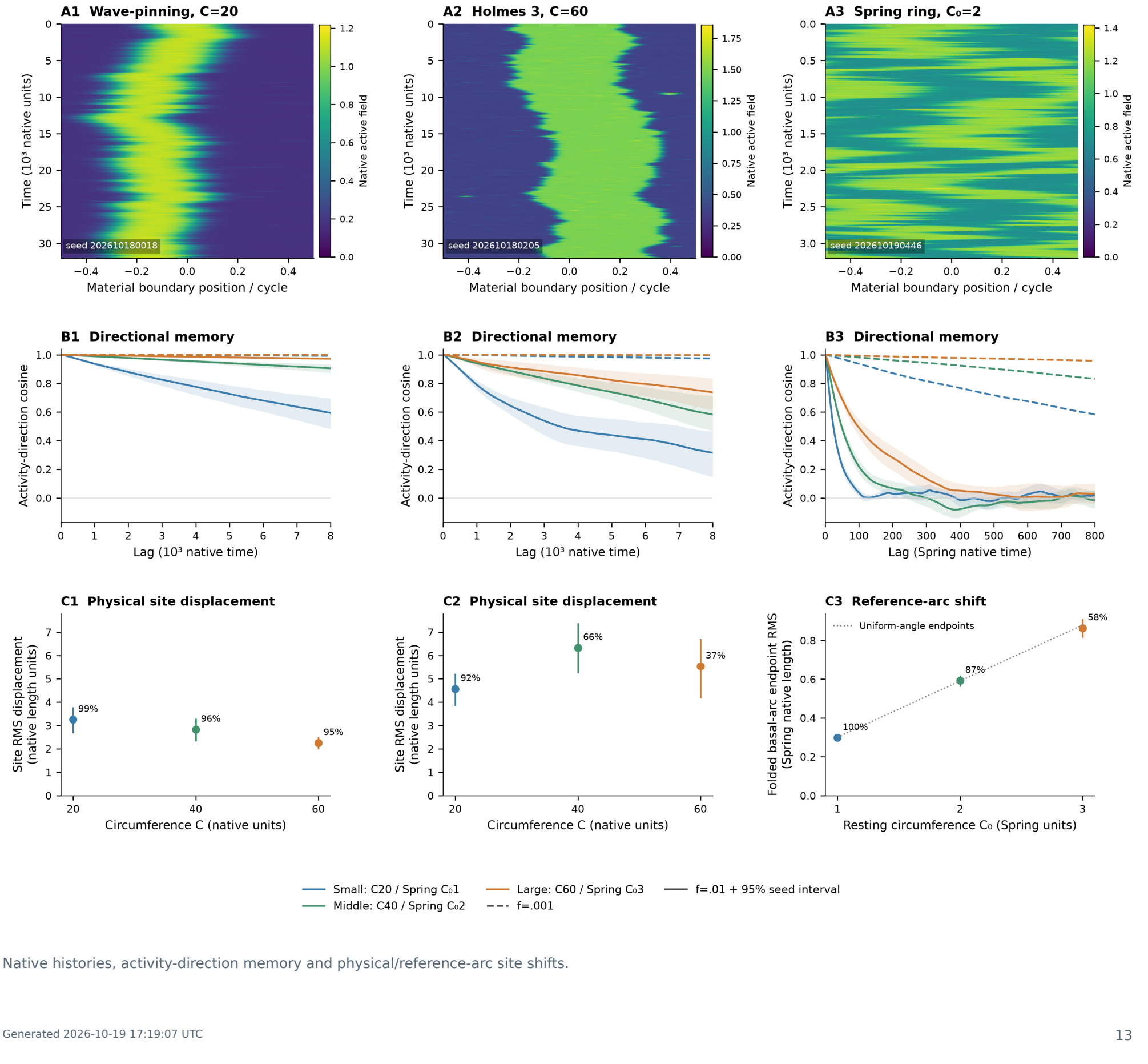
$$M_0 = \sum_j u_j \Delta s_j, \qquad q_1 = \frac{|M_1|}{M_0}, \qquad R(t) = \frac{M_0(t)}{M_0(0)}\frac{q_1(t)}{q_1(0)} \tag{3.3}$$

Relative resultant magnitude combines total activity and directional organization. It can exceed one and differs from current-mass-normalized coherence q1.

Sources and reproducibility

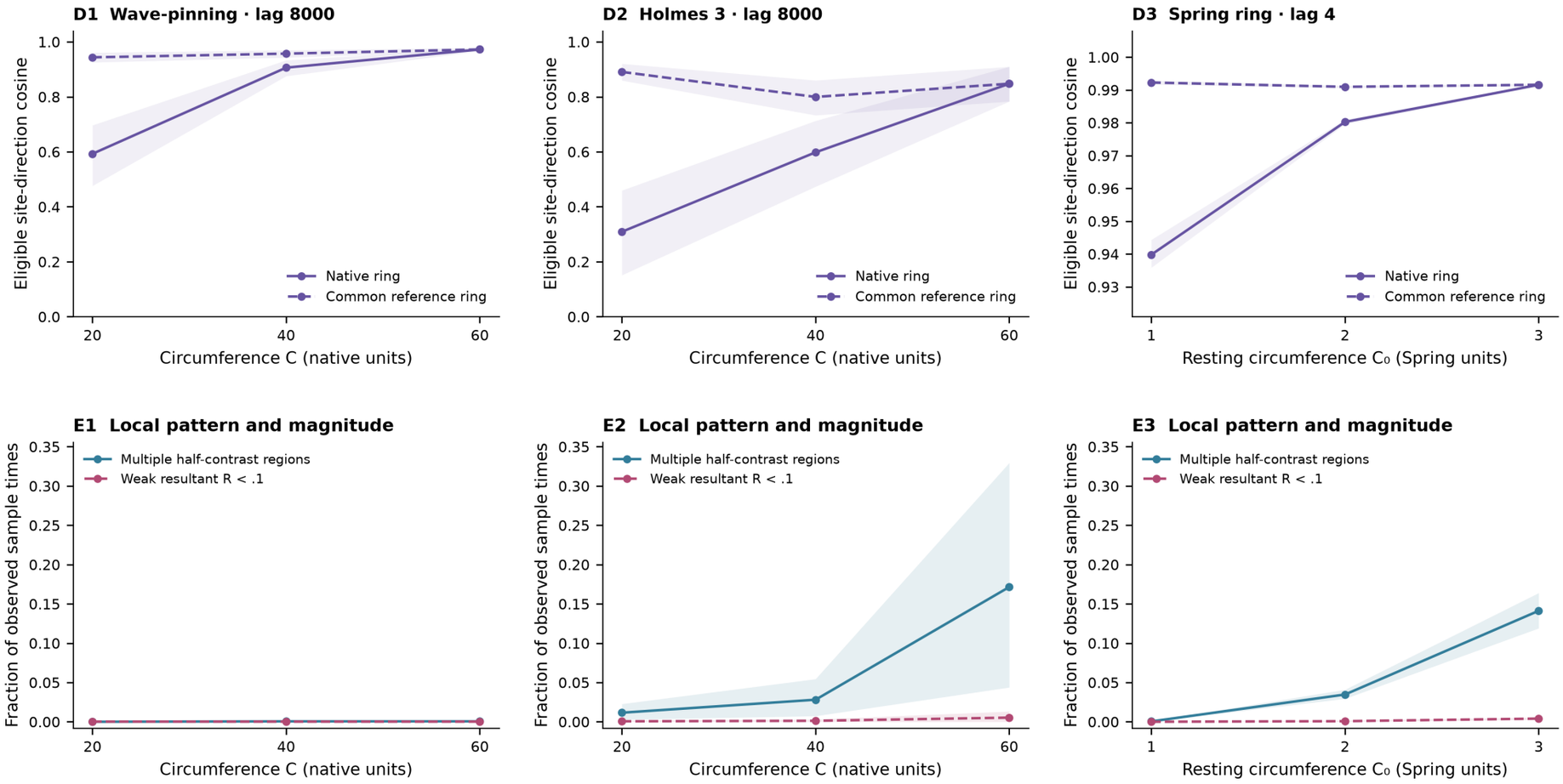
Authoritative complete 240-condition table
Initial profiles, cue footprints and overlaps
Regional crossing brackets and rebound flags
Native archive replay and complete-grid validation
Response-map evidence and enlarged supporting views
Exact physical vector contributions in the teaching example
Controlled placement-contrast evidence
Independent scientific review

Figure 4 · Directional memory and physical site dynamics · image 1 of 2



Native histories, activity-direction memory and physical/reference-arc site shifts.

Figure 4 · Directional memory and physical site dynamics · image 2 of 2



Paired geometric remapping and passive local-pattern/magnitude readouts.

Figure 4 · Directional memory and physical site dynamics

stochastic dynamics

Detailed legend

Figure 4. Directional memory needs a physical site readout. Memory panels B show both predeclared strengths: dashed f=.001, solid f=.01. Histories and site/pattern metrics use f=.01. Every noisy group retains all 12 independent seeds; shading and bars are pointwise 95% whole-seed intervals (B shades the stronger input). Native units differ across models.

A1–A3: native active-field histories from the predeclared condition medoids: Wave-pinning C20, Holmes C60 and Spring $C_0$2. Each model uses a zero-based native activity scale; boundary position is material position around the ring.

B1–B3: whole-window activity-direction cosine, conditional on both endpoint relative first-moment magnitudes R≥.1. The physical activity moment $M_1(t)$ is divided by its initial complex value. Activity direction is distinct from cell movement.

C1–C3: dominant half-contrast-site endpoint RMS shift at lag 8000 for chemical rings or 800 for Spring; percentages are eligible endpoint pairs. Spring uses basal-reference arc shift Cbasal×wrapped material-position change, with the dotted uniform-angle reference Cbasal/√12. Long-lag folded shifts can approach circular mixing and are not travel distances. Spring has its own native length axis.

D1–D3: native site-angle cosine and remapping of the same eligible shortest shifts to common circumference C*=60 for chemical rings or 3.04517 for Spring. Chemical panels use lag 8000; Spring uses early lag 4 before late endpoint mixing. Full lag curves and 4/50/800 comparisons remain supporting evidence. This is a geometric coordinate comparison, not a causal decomposition of global activity-vector memory.

E1–E3: observed time fractions with multiple current half-contrast activity regions and weak relative resultant R<.1. Region multiplicity does not establish functional competition. All 6401 scalar samples per seed enter each fraction. Supporting increments retain rare >π/4 Spring changes at cadence .5 and none in chemical traces at cadence 5, conditional on both endpoint R≥.1; these sampled diagnostics do not define an event class or exclude faster changes.

Complete Goryachev results remain supporting evidence: its high-input C20 mean control also loses polarity, precluding an intact-state wandering interpretation. Both declared strengths, all mean/no-input controls and medoids, magnitude and increment distributions, eligibility sensitivities and full/half-window analyses are retained. Methods define Spring's approved stretch/dilution/diffusion carryover, mean-force global-extension closure and finite transported-material noise law.

Narrative points and conclusions

Figure 4 · Directional memory and physical site dynamics · continued

- Angular-memory and physical site-shift measurements can show different size trends. Spring makes the forcing dependence explicit: weak-input lag-800 direction cosine rises .584/.832/.958 across sizes despite nonmonotone basal-arc RMS, whereas stronger input gives .019/−.017/.028 with substantial mean relative first-moment magnitude remaining and late folded shifts near the uniform-angle reference. All three main models retain their native units and controls.
- Pairwise coordinate remapping exposes the geometric part of site-angle retention, while local-region and magnitude readouts distinguish observable organizations without classifying abrupt events or inferring competition.

Proposed measurements and experimental follow-up

Activity-direction correlation and full-vector memory

Physical/reference-arc site RMS with endpoint coverage

Native/common-reference site-angle retention on identical pairs

Local-region and weak-resultant occupancy

Gated increment distributions and full/half-window sensitivity

Scope, limitations and pending validation

Native units are not calibrated across models.

Activity direction does not assign movement.

Endpoint pairs do not establish site ancestry.

Spring reference-arc shifts separate material change from uniform stretch.

Remapping is a coordinate comparison, not a causal decomposition.

Finite-knot/noise-law and finite-window limitations remain explicit.

Sources and reproducibility

Original chemical-ring stochastic report
Original noise design
Complete Goryachev/chemical interim figure
Frozen Spring main design
Paired finite-law estimator sensitivity
Spring full analysis audit
Spring centroid/half-window/CDF/forcing audit
Spring full seed coverage
Spring methods and limits

Figure 4 · Directional memory and physical site dynamics · continued

Final four-model quantitative evidence
Spring raw full/half-window memory
Spring site and stretch metrics
Spring first/last-half site metrics
Spring magnitude-gated increment distributions
Spring actual forcing audits
Spring early/late endpoint summary
Independent scientific review
Independent requirement and evidence audit (assembly snapshot)
Spring active amount and spatial-balance methods
Spring scalar amount/profile audit
Spring active amount and balance condition summaries
All81 Spring scalar case exports

Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Supplementary material and supporting analyses · continued

Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification